Solutions

Maths

Q1. Let P and Q be the points of contact of the common tangents to the parabola $y^2 = 4x$ and the circle $x^2 + y^2 - 6x = 0$ on the parabola, and let R and S be their corresponding points of contact on the circle. Then the area (in sq. units) of the quadrilateral $PRSQ$ is

(A) $\frac{21\sqrt{3}}{4}$ (B) $\frac{21\sqrt{3}}{2}$ (C) $\frac{7\sqrt{3}}{2}$ (D) $\frac{7\sqrt{3}}{4}$

Ans. A

Sol. The equation of any tangent to the parabola $y^2 = 4x$ is:

$$y = mx + \frac{1}{m} \implies m^2x - my + 1 = 0$$

Since the line is also tangent to the circle, the perpendicular distance from the center $(3, 0)$ to the line must equal the radius:

$$\frac{|3m^2 + 1|}{\sqrt{m^4 + m^2}} = 3 \implies (3m^2 + 1)^2 = 9m^2(m^2 + 1)$$

$$9m^4 + 6m^2 + 1 = 9m^4 + 9m^2 \implies 3m^2 = 1 \implies m = \pm\frac{1}{\sqrt{3}}$$

For $m = \frac{1}{\sqrt{3}}$, the point of contact is $P(3, 2\sqrt{3})$.

For $m = -\frac{1}{\sqrt{3}}$, the point of contact is $Q(3, -2\sqrt{3})$.

For $m = \frac{1}{\sqrt{3}}$, the tangent is $x - \sqrt{3}y + 3 = 0$ and the normal is $\sqrt{3}x + y - 3\sqrt{3} = 0$. Solving these gives $R(\frac{3}{2}, \frac{3\sqrt{3}}{2})$.

For $m = -\frac{1}{\sqrt{3}}$, by symmetry, the point of contact is $S\left(\frac{3}{2}, -\frac{3\sqrt{3}}{2}\right)$.

The vertices of the quadrilateral $PRSQ$ are $P(3, 2\sqrt{3})$, $R\left(\frac{3}{2}, \frac{3\sqrt{3}}{2}\right)$, $S\left(\frac{3}{2}, -\frac{3\sqrt{3}}{2}\right)$ and $Q(3, -2\sqrt{3})$.

The area of the trapezoid $PRSQ$ is:

$$\begin{aligned}\text{Area} &= \frac{1}{2} \times (PQ + RS) \times h \\ &= \frac{1}{2} \times (4\sqrt{3} + 3\sqrt{3}) \times \frac{3}{2} \\ &= \frac{21\sqrt{3}}{4}\end{aligned}$$

Q2. Consider a continuous frequency distribution given below:

Class Interval	Frequency
0-10	3
10-20	4
20-30	3

If μ and σ^2 denote the mean and the variance of this distribution respectively, then $\mu + \sigma^2$ is equal to:

(A) 75 (B) 65 (C) 55 (D) 45

Ans. A

Sol. The midpoints (x_i) of the class intervals $0-10$, $10-20$, and $20-30$ are 5, 15, and 25 respectively.

The corresponding frequencies are $f_1 = 3$, $f_2 = 4$, and $f_3 = 3$. Total frequency $N = \sum f_i = 3 + 4 + 3 = 10$.

The mean μ is calculated as:

$$\begin{aligned}\mu &= \frac{\sum f_i x_i}{N} \\ &= \frac{3(5) + 4(15) + 3(25)}{10} \\ &= \frac{15 + 60 + 75}{10} \\ &= 15\end{aligned}$$

The variance σ^2 is calculated as:

$$\begin{aligned}\sigma^2 &= \frac{\sum f_i (x_i - \mu)^2}{N} \\ &= \frac{3(5-15)^2 + 4(15-15)^2 + 3(25-15)^2}{10}\end{aligned} \quad \sigma^2 = \frac{3(100) + 4(0) + 3(100)}{10} = \frac{600}{10} = 60$$

Thus, the required value is:

$$\mu + \sigma^2 = 15 + 60 = 75$$

Q3. The value of the limit $\lim\limits_{x \to 0} \dfrac{e^{x \cot x} - e}{x^2}$ is:

(A) $-\dfrac{e}{2}$ (B) $-\dfrac{e}{3}$ (C) $\dfrac{e}{3}$ (D) $-\dfrac{1}{3}$

Ans. B

Sol. Applying L'Hospital's Rule

$$\lim_{x \to 0} \frac{e^{x \cot x}(\cot x - x \csc^2 x)}{2x}$$

Let's rewrite the trigonometric expression in terms of sine and cosine:

$$\lim_{x\to 0}\frac{\cot x - x\csc^2 x}{2x} = \lim_{x\to 0}\frac{\frac{\cos x}{\sin x} - \frac{x}{\sin^2 x}}{2x}$$
$$= \lim_{x\to 0}\frac{\sin x\cos x - x}{2x\sin^2 x}$$
$$\lim_{x\to 0}\frac{\frac{1}{2}\sin 2x - x}{2x^3}$$
Applying L'Hospital's Rule again (form $\frac{0}{0}$):
$$\lim_{x\to 0}\frac{\cos 2x - 1}{6x^2}$$
Applying L'Hospital's Rule one more time:
$$\lim_{x\to 0}\frac{-2\sin 2x}{12x} = -\frac{4}{12} = -\frac{1}{3}$$
Combining this with the front factor e, the required limit is:
$$\text{Limit} = e\times\left(-\frac{1}{3}\right) = -\frac{e}{3}$$

Q4. Consider a continuous frequency distribution with class intervals $0-2$, $2-4$, and $4-6$ having frequencies f_1, 6, and f_3 respectively. If the total frequency is 10 and the mean of the distribution is 3, let the mean deviation about the mean of this distribution be p. If a random variable X follows a binomial distribution $B(n,p)$ with $n=4$, then the probability $P(X\geq 3)$ is equal to:

(A) $\frac{256}{625}$ (B) $\frac{512}{625}$ (C) $\frac{128}{625}$ (D) $\frac{432}{625}$

Ans. B

Sol. Given total frequency:

$$\sum f_i = f_1 + 6 + f_3 = 10 \implies f_1 + f_3 = 4$$

The midpoints of the class intervals are $x_1 = 1$, $x_2 = 3$, $x_3 = 5$.

Given mean ($\bar{x}$) = 3:

$$\bar{x} = \frac{f_1(1) + 6(3) + f_3(5)}{10} = 3 \implies f_1 + 5f_3 = 12$$

Solving the two equations:

$$f_3 = 2, \quad f_1 = 2$$

The mean deviation about the mean (p) is given by:

$$p = \frac{\sum f_i|x_i - \bar{x}|}{\sum f_i}$$
$$= \frac{2|1-3| + 6|3-3| + 2|5-3|}{10}$$
$$= \frac{4+0+4}{10}$$
$$= \frac{4}{5}$$

For the binomial distribution $B(4, 4/5)$, the probability $P(X\geq 3)$ is:

$$P(X\geq 3) = P(X=3) + P(X=4)$$

$$P(X \geq 3)$$
$$= \binom{4}{3}\left(\frac{4}{5}\right)^3\left(\frac{1}{5}\right)^1 + \binom{4}{4}\left(\frac{4}{5}\right)^4$$
$$P(X \geq 3)$$
$$= 4\left(\frac{64}{625}\right) + \frac{256}{625}$$
$$= \frac{256+256}{625}$$
$$= \frac{512}{625}$$

Q5. Let L_1 be the line $y=0$ and L_2 be a line passing through the origin. Let the line $L_3 : x - \sqrt{3}y = 0$ be the internal angle bisector of the angle between L_1 and L_2. If $P(2,0)$ is a point on L_1, then the coordinates of the image of point P with respect to the line L_3 are:

(A) $\left(1, \sqrt{3}\right)$ (B) $\left(\sqrt{3}, 1\right)$ (C) $\left(1, -\sqrt{3}\right)$ (D) $\left(-1, \sqrt{3}\right)$

Ans. A

Sol. The image (x, y) of a point (x_1, y_1) with respect to the line $ax + by + c = 0$ can be determined using the standard transformation formula:

$$\frac{x - x_1}{a} = \frac{y - y_1}{b} = -2\frac{ax_1 + by_1 + c}{a^2 + b^2}$$

Here, the given point is $P(2,0)$ and the line is $x - \sqrt{3}y = 0$.

Thus, the coordinates of the image of point P are $\left(1, \sqrt{3}\right)$.

Alternative Method:

By the geometric property of angle bisectors, the image of any point lying on one of the lines with respect to their angle bisector always lies on the other line, maintaining the same distance from the vertex (the intersection point). The vertex here is the origin (0, 0). The distance of P(2, 0) from the origin is 2 units. Since the line L_3 makes an angle of 30° with the x-axis (L_1), the line L_2 must make an angle of 60° with the positive x-axis.

The image point lies on L_2 at a distance of 2 units from the origin, which gives its parametric coordinates as :

(2 cos60°, 2 sin60°) = $\left(1, \sqrt{3}\right)$

Q6. Consider the limit $L = \lim\limits_{x \to 0}\left(\frac{e^x \cos x - 1}{x}\right)^{\frac{1}{x^2}}$ Then the value of L is equal to:

(A) $e^{-\frac{1}{2}}$ (B) $e^{-\frac{1}{3}}$ (C) $e^{-\frac{1}{6}}$ (D) 1

Ans. B

Sol. The given limit is of the form 1^∞ since $\lim\limits_{x \to 0}\frac{e^x \cos x - 1}{x} = 1$ and $\lim\limits_{x \to 0}\frac{1}{x^2} = \infty$.

Using the property of 1^∞ form, we can write:

$L = e^K$

where $K = \lim\limits_{x \to 0}\frac{1}{x^2}\left(\frac{e^x \cos x - 1}{x} - 1\right) = \lim\limits_{x \to 0}\frac{e^x \cos x - 1 - x}{x^3}$

Using the Maclaurin series expansions for e^x and $\cos x$:

$$e^x \cos x = \left(1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \ldots\right)\left(1 - \frac{x^2}{2} + \ldots\right)$$

$$e^x \cos x = 1 + x - \frac{x^3}{3} + \ldots$$

Substitute this expansion back into the expression for K:

$$K = \lim_{x \to 0} \frac{\left(1 + x - \frac{x^3}{3} + \ldots\right) - 1 - x}{x^3}$$

$$K = \lim_{x \to 0} \frac{-\frac{x^3}{3} + \ldots}{x^3} = -\frac{1}{3}$$

Hence, the value of the limit is:

$L = e^{-\frac{1}{3}}$

Q7. Let $f : \mathbb{R} \to \mathbb{R}$ be a function defined by: $f(x) = \begin{cases} x^2 + a, & x \leq 1 \\ bx + 2, & x > 1 \end{cases}$ If $f(x)$ is differentiable at $x = 1$, then the value of $a + b$ is:

(A) 5 (B) 3 (C) 4 (D) 2

Ans. A

Sol. Since $f(x)$ is differentiable at $x = 1$, it must be continuous at $x = 1$.

$\lim\limits_{x \to 1^-} f(x) = \lim\limits_{x \to 1^+} f(x) \implies 1 + a = b + 2 \implies a - b = 1$

Equating the left-hand derivative and right-hand derivative at $x = 1$:

$$\frac{d}{dx}(x^2 + a)\Big|_{x=1} = \frac{d}{dx}(bx + 2)\Big|_{x=1}$$

$$\implies 2$$

$$= b$$

Substituting $b = 2$ into the continuity relation:

$a - 2 = 1 \implies a = 3$

Therefore, $a + b = 3 + 2 = 5$.

Q8. For $x \in (0, \pi/2)$, evaluate the indefinite integral: $\displaystyle\int \frac{\sin x - \cos x}{(\sin x + \cos x)\sqrt{\sin 2x}}\,dx$

(A) $\sec^{-1}(\sin x + \cos x) + C$ (B) $-\sec^{-1}(\sin x + \cos x) + C$ (C) $\sec^{-1}(\sin x - \cos x) + C$

(D) $-\sec^{-1}(\sin x - \cos x) + C$

Ans. B

Sol. Let the given integral be I: $I = \displaystyle\int \frac{\sin x - \cos x}{(\sin x + \cos x)\sqrt{\sin 2x}}\,dx$

Substitute $\sin x + \cos x = t$.

$(\cos x - \sin x)dx = dt \implies (\sin x - \cos x)dx = -dt$

Squaring both sides of the substitution:

$(\sin x + \cos x)^2 = t^2 \implies 1 + \sin 2x = t^2 \implies \sin 2x = t^2 - 1$

Substituting these into the integral I: $I = \displaystyle\int \frac{-dt}{t\sqrt{t^2 - 1}}$

Since $x \in (0, \pi/2)$, we have $t = \sin x + \cos x > 1$. Evaluating the standard integral gives:

$I = -\sec^{-1}(t) + C = -\sec^{-1}(\sin x + \cos x) + C$

Q9. If a curve is given parametrically by the equations $x = a(\theta + \sin\theta)$ and $y = a(1 - \cos\theta)$, where $a > 0$, then the value of $\frac{dy}{dx}$ at $\theta = \frac{\pi}{2}$ is equal to:

(A) 1 (B) -1 (C) 0 (D) $\frac{1}{2}$

Ans. A

Sol. Given the parametric equations of the curve:

$x = a(\theta + \sin\theta)$

$y = a(1 - \cos\theta)$

Differentiating both equations with respect to the parameter θ:

$$\frac{dx}{d\theta} = a(1 + \cos\theta)$$

$$\frac{dy}{d\theta} = a\sin\theta$$

$$\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}}$$

$$= \frac{a\sin\theta}{a(1+\cos\theta)}$$

$$= \frac{\sin\theta}{1+\cos\theta}$$

$$\left.\frac{dy}{dx}\right|_{\theta=\frac{\pi}{2}} = \frac{\sin(\pi/2)}{1+\cos(\pi/2)}$$

$$= \frac{1}{1+0}$$

$$= 1$$

Q10. The value of x at which the function $f(x) = 2x^3 - 9x^2 + 12x + 1$ attains its local minimum is

(A) 1 (B) 2 (C) 3 (D) 0

Ans. B

Sol. Given function is $f(x) = 2x^3 - 9x^2 + 12x + 1$.

Differentiating with respect to x:

$f'(x) = 6x^2 - 18x + 12 = 6(x-1)(x-2)$

Setting $f'(x) = 0$ gives the critical points $x = 1$ and $x = 2$.

Differentiating again:

$f''(x) = 12x - 18$

Evaluating at the critical points:

For $x = 1$: $f''(1) = -6 < 0$, which signifies a point of local maximum.

For $x = 2$: $f''(2) = 6 > 0$, which signifies a point of local minimum.

Q11. Let $f(x) = \frac{x}{2} + \frac{a}{x}$. The complete set of real values of a for which $f(x)$ is monotonic in the interval $(1, 2)$ is:

(A) $\left(-\infty, \frac{1}{2}\right] \cup [2, \infty)$ (B) $\left[\frac{1}{2}, 2\right]$ (C) $\left(-\infty, \frac{1}{2}\right]$ (D) $[2, \infty)$

Ans. A

Sol. Given function is $f(x) = \frac{x}{2} + \frac{a}{x}$.

Differentiating with respect to x:

$f'(x) = \frac{1}{2} - \frac{a}{x^2} = \frac{x^2 - 2a}{2x^2}$

For $f(x)$ to be monotonic in $(1, 2)$, $f'(x)$ must not change its sign in the interval $(1, 2)$. Since the denominator $2x^2 > 0$ for all $x \in (1, 2)$, the sign of $f'(x)$ is entirely determined by the numerator $x^2 - 2a$.

Case I: $f(x)$ is non-decreasing in $(1, 2)$

$\implies f'(x) \geq 0 \implies x^2 - 2a \geq 0 \implies 2a \leq x^2 \quad \forall x \in (1, 2)$

$\implies 2a \leq \min\limits_{x \in (1,2)} (x^2) = 1 \implies a \leq \frac{1}{2}$

Case II: $f(x)$ is non-increasing in $(1, 2)$

$\implies f'(x) \leq 0 \implies x^2 - 2a \leq 0 \implies 2a \geq x^2 \quad \forall x \in (1, 2)$

$\implies 2a \geq \max_{x \in (1,2)}(x^2) = 4 \implies a \geq 2$

Combining both cases, the complete set of values for a is $\left(-\infty, \frac{1}{2}\right] \cup [2, \infty)$.

Q12. Let $f(x) = \sin(x^2)$ and $g(x) = 1 - \cos x$. If the derivative of $f(x)$ with respect to $g(x)$ is represented by $h(x)$, then the value of $\lim\limits_{x \to 0} \frac{h(x) - 2}{x^2}$ is equal to:

(A) $\frac{1}{6}$ (B) $\frac{1}{3}$ (C) $\frac{2}{3}$ (D) 1

Ans. B

Sol. The derivative of $f(x)$ with respect to $g(x)$ is given by:

$h(x) = \frac{f'(x)}{g'(x)} = \frac{2x \cos(x^2)}{\sin x}$

We need to evaluate the limit:

$L = \lim\limits_{x \to 0} \frac{\frac{2x \cos(x^2)}{\sin x} - 2}{x^2}$

$= \lim\limits_{x \to 0} \frac{2x \cos(x^2) - 2 \sin x}{x^2 \sin x}$

Since $\lim\limits_{x \to 0} \frac{\sin x}{x} = 1$, we can rewrite the limit as:

$L = \lim\limits_{x \to 0} \frac{2x \cos(x^2) - 2 \sin x}{x^3}$

This is a $\frac{0}{0}$ indeterminate form. Applying L'Hospital's Rule:

$L = \lim\limits_{x \to 0} \frac{2 \cos(x^2) - 4x^2 \sin(x^2) - 2 \cos x}{3x^2}$

This is again a $\frac{0}{0}$ form. Applying L'Hospital's Rule a second time:

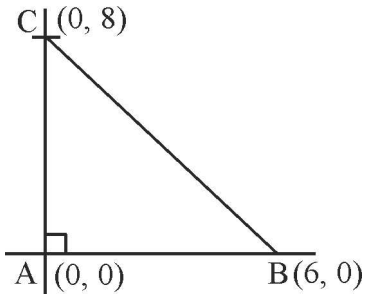
$$L = \lim_{x \to 0} \frac{-4x\sin(x^2) - 8x\sin(x^2) - 8x^3\cos(x^2) + 2\sin x}{6x}$$
$$L = \lim_{x \to 0} \frac{-12x\sin(x^2) - 8x^3\cos(x^2) + 2\sin x}{6x}$$
$$L = \lim_{x \to 0} \left(-2\sin(x^2) - \frac{4}{3}x^2\cos(x^2) + \frac{1}{3} \cdot \frac{\sin x}{x}\right)$$
$$L = 0 - 0 + \frac{1}{3}(1) = \frac{1}{3}$$

Q13. Let the vertices of a triangle be $A(0,0)$, $B(6,0)$ and $C(0,8)$. If G and H denote the centroid and the orthocenter of $\triangle ABC$ respectively, then the distance GH is equal to:

(A) $\frac{5}{3}$ (B) 5 (C) $\frac{10}{3}$ (D) $\frac{20}{3}$

Ans. C

Sol.



H(0, 0)

$G\left(2, \frac{8}{3}\right)$

$HG = \frac{10}{3}$

Q14. A committee of 4 members is to be selected at random from a group consisting of 5 men and 4 women. The probability that the selected committee contains an equal number of men and women is

(A) $\frac{5}{14}$ (B) $\frac{10}{21}$ (C) $\frac{2}{7}$ (D) $\frac{4}{9}$

Ans. B

Sol. The total number of ways to select 4 members from a total of 9 individuals (5 men and 4 women) is given by the total number of outcomes in the sample space S:

$$n(S) = \binom{9}{4}$$
$$= \frac{9 \times 8 \times 7 \times 6}{4 \times 3 \times 2 \times 1}$$
$$= 126$$

The favorable event E requires the committee to have an equal number of men and women, which means selecting exactly 2 men and 2 women:

$$n(E) = \binom{5}{2} \times \binom{4}{2} = 10 \times 6 = 60$$

The required probability is:

$$P(E) = \frac{n(E)}{n(S)} = \frac{60}{126} = \frac{10}{21}$$

Q15. An unbiased die is rolled 3 times. A success is defined as obtaining a prime number on a roll. If the die shows a success at least once, then the conditional probability of obtaining exactly two successes is:

(A) $\frac{3}{7}$ (B) $\frac{1}{3}$ (C) $\frac{3}{8}$ (D) $\frac{1}{2}$

Ans. A

Sol. Let X be the random variable representing the number of successes in $n = 3$ trials.

The probability of success in a single trial (rolling a prime number from $2, 3, 5$) is $p = \frac{3}{6} = \frac{1}{2}$ and the probability of failure is $q = 1 - p = \frac{1}{2}$.

The random variable X follows a binomial distribution. We are required to evaluate the conditional probability $P(X = 2|X \geqslant 1)$.

By the definition of conditional probability:

$P(X = 2|X \geqslant 1) = \frac{P(X = 2 \cap X \geqslant 1)}{P(X \geqslant 1)} = \frac{P(X = 2)}{P(X \geqslant 1)}$

Using the binomial probability mass function $P(X = r) = \binom{n}{r} p^r q^{n-r}$:

$P(X = 2) = \binom{3}{2}\left(\frac{1}{2}\right)^2\left(\frac{1}{2}\right)^1 = \frac{3}{8}$

$P(X \geqslant 1) = 1 - P(X = 0) = 1 - \binom{3}{0}\left(\frac{1}{2}\right)^3 = 1 - \frac{1}{8} = \frac{7}{8}$

$P(X = 2|X \geqslant 1) = \frac{\frac{3}{8}}{\frac{7}{8}} = \frac{3}{7}$

Q16. Let $f : \mathbb{R} \to \mathbb{R}$ be a function defined by $f(x) = x^3 + 3x^2 + ax + 7$, where a is a real constant. If $f(x)$ is strictly increasing for all $x \in \mathbb{R}$, then the set of all possible values of a is:

(A) $[3, \infty)$ (B) $(-\infty, 3]$ (C) $(3, \infty)$ (D) $[9, \infty)$

Ans. A

Sol. Given function is: $f(x) = x^3 + 3x^2 + ax + 7$

Differentiating with respect to x: $f'(x) = 3x^2 + 6x + a$

For $f(x)$ to be strictly increasing for all $x \in \mathbb{R}$, the derivative $f'(x)$ must be non-negative for all $x \in \mathbb{R}$.

Since the coefficient of x^2 is $3 > 0$, the quadratic expression $3x^2 + 6x + a \geq 0$ holds true for all $x \in \mathbb{R}$ if its discriminant $D \leq 0$:

$D = 6^2 - 4(3)(a) \leq 0$ $36 - 12a \leq 0$ $12a \geq 36 \implies a \geq 3$

Thus, $a \in [3, \infty)$.

Q17. The value of the limit

$\lim\limits_{x \to 0}\left([x] + \frac{\sin x}{\{x\}}\right)^2$ where $[x]$ denotes the greatest integer function and $\{x\}$ denotes the

fractional part function, is equal to:

(A) 0 (B) 1 (C) -1 (D) does not exist

Ans. B

Sol. To evaluate the given limit, we need to determine the Right-Hand Limit (RHL) and the Left-Hand Limit (LHL) separately at $x = 0$.

1. Right-Hand Limit (RHL):

As $x \to 0^+$, x is a small positive number.

Therefore, $[x] = 0$ and $\{x\} = x - [x] = x$.

Substituting these values into the expression:

$$\text{RHL} = \lim_{x \to 0^+} \left(0 + \frac{\sin x}{x}\right)^2$$

Since $\lim\limits_{x \to 0} \dfrac{\sin x}{x} = 1$:

$$\text{RHL} = (1)^2 = 1$$

2. Left-Hand Limit (LHL):

As $x \to 0^-$, x is a small negative number.

Therefore, $[x] = -1$ and $\{x\} = x - [x] = x + 1$.

Substituting these values into the expression:

$$\text{LHL} = \lim_{x \to 0^-} \left(-1 + \frac{\sin x}{x+1}\right)^2$$

As $x \to 0^-$, $\sin x \to 0$ and $x + 1 \to 1$, which gives $\dfrac{\sin x}{x+1} \to 0$.

$$\text{LHL} = (-1 + 0)^2 = 1$$

Since $\text{RHL} = \text{LHL} = 1$, the limit exists and its value is equal to 1.

Q18. Let $A(x)$ denote the area of a rectangle inscribed in a semicircle of fixed radius R, where x represents half of the length of the rectangle's base lying along the diameter of the semicircle. Let x_1 be the value of x for which the area of the rectangle is maximized. If Lagrange's Mean Value Theorem is applied to the function $A(x)$ on the interval $[0, x_1]$, then the value of $c \in (0, x_1)$ that satisfies $A'(c) = \dfrac{A(x_1) - A(0)}{x_1 - 0}$ is:

(A) $\dfrac{R}{2}$ (B) $\dfrac{R}{2\sqrt{2}}$ (C) $\dfrac{R}{4}$ (D) None of these

Ans. D

Sol. The dimensions of the inscribed rectangle are base length $2x$ and height $\sqrt{R^2 - x^2}$. Thus, the area function $A(x)$ for $x \in [0, R]$ is given by:

$A(x) = 2x\sqrt{R^2 - x^2}$

Differentiating $A(x)$ with respect to x:

$$A'(x) = 2\sqrt{R^2 - x^2} - \frac{2x^2}{\sqrt{R^2 - x^2}}$$

$$= \frac{2(R^2 - 2x^2)}{\sqrt{R^2 - x^2}}$$

To maximize the area, we set $A'(x) = 0$:

$$R^2 - 2x^2 = 0 \implies x_1 = \frac{R}{\sqrt{2}}$$

Q19. The indefinite integral $\int \frac{\sqrt{1-x^2}}{x^4}dx$ is equal to (where C is the constant of integration):

(A) $-\frac{(1-x^2)^{3/2}}{3x^3} + C$ (B) $-\frac{(1-x^2)^{3/2}}{3x^2} + C$ (C) $\frac{(1-x^2)^{3/2}}{3x^3} + C$ (D) $-\frac{\sqrt{1-x^2}}{3x^3} + C$

Ans. A

Sol. Let the given integral be $I = \int \frac{\sqrt{1-x^2}}{x^4}dx$.

Substitute $x = \sin\theta \implies dx = \cos\theta d\theta$.

$$I = \int \frac{\sqrt{1-\sin^2\theta}}{\sin^4\theta}\cos\theta d\theta$$
$$= \int \frac{\cos^2\theta}{\sin^4\theta}d\theta$$
$$= \int \cot^2\theta \csc^2\theta d\theta$$

Now, put $\cot\theta = t \implies \csc^2\theta d\theta = -dt$.

$$I = \int -t^2 dt = -\frac{t^3}{3} + C = -\frac{\cot^3\theta}{3} + C$$

Since $\cot\theta = \frac{\sqrt{1-x^2}}{x}$, we get:

$$I = -\frac{(1-x^2)^{3/2}}{3x^3} + C$$

Q20. If the curves $y = x^2 + k$ and $y = 2 - x^2$ intersect orthogonally at their points of intersection, then the value of k is:

(A) 1 (B) $\frac{3}{2}$ (C) 2 (D) $\frac{1}{2}$

Ans. B

Sol. Let (x, y) be the point of intersection of the two curves. Equating the expressions for y:

$x^2 + k = 2 - x^2 2x^2 = 2 - k \implies x^2 = \frac{2-k}{2}$

Differentiating both equations with respect to x to find the slopes of the tangents (m_1 and m_2):

For the first curve: $m_1 = \frac{dy}{dx} = 2x$

For the second curve: $m_2 = \frac{dy}{dx} = -2x$

$m_1 \cdot m_2 = -1(2x)(-2x) = -1 \implies -4x^2 = -1 \implies x^2 = \frac{1}{4}$

Equating the two expressions obtained for x^2:

$\frac{2-k}{2} = \frac{1}{4} 2 - k = \frac{1}{2} k = 2 - \frac{1}{2} = \frac{3}{2}$

Q1. Let $f:[-1,1]\to\mathbb{R}$ be a continuous function satisfying $f(-1)=4$ and $f(1)=-2$. Let $g(x)=f(x)-x^4-2$. Find the minimum number of distinct real roots of the equation $f(x)g(x)=0$ in the interval $[-1,1]$.

Ans. 2

Sol. Since $f(x)$ is continuous on $[-1,1]$ with $f(-1)=4>0$ and $f(1)=-2<0$, by the Intermediate Value Theorem (IVT), there exists at least one real number $c_1\in(-1,1)$ such that:

$f(c_1)=0$

Now, consider the function $g(x)=f(x)-x^4-2$. Since $f(x)$ and $-x^4-2$ are continuous on $[-1,1]$, $g(x)$ is also continuous on $[-1,1]$.

Evaluating $g(x)$ at the endpoints:

$g(-1)=f(-1)-(-1)^4-2=4-1-2=1>0$

$g(1)=f(1)-(1)^4-2=-2-1-2=-5<0$

By the Intermediate Value Theorem, there exists at least one real number $c_2\in(-1,1)$ such that:

$g(c_2)=0$

To check if these roots can be identical, suppose $c_1=c_2=c$. Then $f(c)=0$ and $g(c)=0$. Substituting $f(c)=0$ into the definition of $g(x)$ gives:

$g(c)=0-c^4-2=-(c^4+2)$

For any real number c, $c^4+2\geq 2$, which implies $g(c)\leq -2<0$. This contradicts $g(c)=0$. Thus, $c_1\neq c_2$, meaning $f(x)=0$ and $g(x)=0$ cannot share any common real roots.

Therefore, the equation $f(x)g(x)=0$ has at least $1+1=2$ distinct real roots in the interval $[-1,1]$.

Q2. The origin of a rectangular coordinate system is shifted to the point $(1,2)$ and then the coordinate axes are rotated through an angle of 45° in the anticlockwise direction. If the points A and B have coordinates $(2,5)$ and $(-1,6)$ respectively in the original coordinate system, and they lie on the opposite sides of the line $\lambda X-Y+\sqrt{2}=0$ (where X and Y are the coordinates in the new system), then the number of integral values of λ is equal to _____.

Ans. 1

Sol. Let (x,y) be the coordinates of any point in the original system and (X,Y) be its coordinates in the new system. The transformation equations when the origin is shifted to $(h,k)=(1,2)$ and axes are rotated by $\theta=45^\circ$ are given by:

$$x=h+X\cos\theta-Y\sin\theta=1+\frac{X-Y}{\sqrt{2}}$$

$$y=k+X\sin\theta+Y\cos\theta=2+\frac{X+Y}{\sqrt{2}}$$

For point $A(2,5)$: the new coordinates of A as $(2\sqrt{2},\sqrt{2})$.

For point $B(-1,6)$: the new coordinates of B as $(\sqrt{2},3\sqrt{2})$.

The given line in the new system is $L(X,Y) = \lambda X - Y + \sqrt{2} = 0$. Since the points A and B lie on the opposite sides of the line, we must have $L(2\sqrt{2}, \sqrt{2}) \cdot L(\sqrt{2}, 3\sqrt{2}) < 0$.

$$\begin{aligned} L(2\sqrt{2}, \sqrt{2}) &= \lambda(2\sqrt{2}) - \sqrt{2} + \sqrt{2} \\ &= 2\sqrt{2}\lambda \end{aligned}$$

$$\begin{aligned} L(\sqrt{2}, 3\sqrt{2}) &= \lambda(\sqrt{2}) - 3\sqrt{2} + \sqrt{2} \\ &= \sqrt{2}(\lambda - 2) \end{aligned}$$

Thus,

$$\begin{aligned} (2\sqrt{2}\lambda) \cdot (\sqrt{2}(\lambda - 2)) < 0 &\implies 4\lambda(\lambda - 2) < 0 \\ &\implies 0 < \lambda < 2 \end{aligned}$$

The only integral value of λ in the interval $(0, 2)$ is $\lambda = 1$.

Q3. If the image of the line $3x - y + 1 = 0$ with respect to the line $x + y - 2 = 0$ is the line $ax + by + 5 = 0$, then find the value of $a^2 + b^2$.

Ans. 10

Sol. Let the given line be $L_1 : 3x - y + 1 = 0$ and the mirror line be $L_2 : x + y - 2 = 0$.

First, find the point of intersection I of L_1 and L_2.

Substituting $y = 3x + 1$ into L_2:

$$x + (3x + 1) - 2 = 0 \implies 4x = 1 \implies x = \frac{1}{4}$$

Correspondingly, $y = 2 - \dfrac{1}{4} = \dfrac{7}{4}$.

Thus, the point of intersection is $I\left(\dfrac{1}{4}, \dfrac{7}{4}\right)$.

Next, choose a convenient point on L_1, say $A(0, 1)$. Let its image with respect to L_2 be $A'(x', y')$.

Using the image formula:

$$\frac{x' - 0}{1} = \frac{y' - 1}{1} = -2\frac{0 + 1 - 2}{1^2 + 1^2} = 1$$

This yields $x' = 1$ and $y' = 2$. So, $A' = (1, 2)$.

The image line L_3 passes through both $I\left(\dfrac{1}{4}, \dfrac{7}{4}\right)$ and $A'(1, 2)$.

The slope m of L_3 is:

$$\begin{aligned} m &= \frac{2 - \frac{7}{4}}{1 - \frac{1}{4}} \\ &= \frac{\frac{1}{4}}{\frac{3}{4}} \\ &= \frac{1}{3} \end{aligned}$$

The equation of L_3 is:

$$y - 2 = \frac{1}{3}(x - 1) \implies x - 3y + 5 = 0$$

Comparing $x - 3y + 5 = 0$ with the given form $ax + by + 5 = 0$, we get $a = 1$ and $b = -3$.

Therefore:

$$a^2 + b^2 = 1^2 + (-3)^2 = 10$$

Q4. A straight line $x+y=k$ intersects the curve $x^2+y^2-4x-6y+6=0$ at two distinct points A and B. If the lines joining the origin to the points A and B are perpendicular to each other, then the sum of all possible values of k is equal to _____.

Ans. 5

Sol. The equation of the given curve is:

$x^2+y^2-4x-6y+6=0$

The equation of the line is $x+y=k$, which can be written as:

$\frac{x+y}{k}=1$

Homogenising the equation of the curve with the help of the line equation to find the pair of straight lines joining the origin to the points of intersection A and B:

$x^2+y^2-4x\left(\frac{x+y}{k}\right)-6y\left(\frac{x+y}{k}\right)+6\left(\frac{x+y}{k}\right)^2=0$

$k^2(x^2+y^2)-4kx(x+y)-6ky(x+y)+6(x+y)^2=0$

$(k^2-4k+6)x^2+(12-10k)xy+(k^2-6k+6)y^2=0$

the sum of the coefficients of x^2 and y^2 must be equal to zero:

$(k^2-4k+6)+(k^2-6k+6)=02k^2-10k+12=0k^2-5k+6=0$

Solving for k gives:

$k=2$ or $k=3$

Both values satisfy the condition for intersecting the curve at two distinct points (since the perpendicular distance from the center $(2,3)$ to the line is less than the radius $\sqrt{7}$).

Thus, the sum of all possible values of k is $2+3=5$.

Q5. Three urns U_1, U_2 and U_3 contain balls as follows:
Urn U_1 contains 2 white and 3 black balls,
Urn U_2 contains 3 white and 2 black balls, and
Urn U_3 contains 4 white and 1 black ball.
One urn is chosen at random and two balls are drawn from it at random without replacement. If both the drawn balls are found to be white, then the probability that they were drawn from Urn U_3 is

Ans. 0.6, 1

Sol. Let E_1, E_2 and E_3 be the events of selecting the urns U_1, U_2 and U_3 respectively. Since the urns are chosen at random:

$P(E_1)=P(E_2)=P(E_3)=\frac{1}{3}$

Let W be the event of drawing two white balls. The conditional probabilities of drawing two white balls from each urn are:

From Urn U_1: $P(W|E_1)=\frac{\binom{2}{2}}{\binom{5}{2}}=\frac{1}{10}$

From Urn U_2: $P(W|E_2) = \frac{\binom{3}{2}}{\binom{5}{2}} = \frac{3}{10}$

From Urn U_3: $P(W|E_3) = \frac{\binom{4}{2}}{\binom{5}{2}} = \frac{6}{10}$

By Bayes' Theorem, the required probability $P(E_3|W)$ is given by:

$$P(E_3|W) = \frac{P(E_3) \cdot P(W|E_3)}{P(E_1) \cdot P(W|E_1) + P(E_2) \cdot P(W|E_2) + P(E_3) \cdot P(W|E_3)}$$

$$P(E_3|W) = \frac{\frac{1}{3} \times \frac{6}{10}}{\left(\frac{1}{3} \times \frac{1}{10}\right) + \left(\frac{1}{3} \times \frac{3}{10}\right) + \left(\frac{1}{3} \times \frac{6}{10}\right)}$$

$$P(E_3|W) = \frac{6}{1+3+6} = \frac{6}{10} = 0.6$$